Ben Motz

1101 East 10th Street
Bloomington, Indiana 47408

bmotz@iu.edu
(812)855-0318

ORCID ID: 0000-0002-0379-2184

Education

2018 **Indiana University – Bloomington, Indiana**
PhD in Cognitive Science
Minor in Psychology

2005 **University of California San Diego – La Jolla, California**
MS in Cognitive Science

2002 **Indiana University – Bloomington, Indiana**
BS in Cognitive Science
Concentrations in Neuroscience and Computer Science
University & Departmental Honors

Professional Experience

August 2022 – *current* **Assistant Professor**, Department of Psychological and Brain Sciences
July 2018 – July 2022 **Research Scientist**, Department of Psychological and Brain Sciences
August 2008 – June 2018 **Senior Lecturer**, Department of Psychological and Brain Sciences
Indiana University (Bloomington, Indiana)

January 2006 – June 2008 **Statistical Consultant**, Analytics
Rapp Collins Worldwide (currently RAPP; El Segundo, California)

September 2003 – January 2006 **Graduate Associate Instructor**, Department of Cognitive Science
University of California San Diego (La Jolla, California)

May 2001 – August 2003 **Research Assistant**, Department of Psychological and Brain Sciences
Indiana University (Bloomington, Indiana)

Grants

Gates Foundation. "Planning Grant for Efficacy Research on AI Innovations" Role: PI. $140,650 2026-2027

National Science Foundation (NSF), "Planning: AI-Ready: TOPSAIL: The AI-Ready Test Bed in Education" Role: PI (Co-PIs: Beth Plale, Danielle McNamara, Mary Styers). $199,990 2025-2027

Tools Competition (2023-2024), Learning Sciences Research Track "Terracotta Messages" Role: PI. $300,000 2024

Institute of Education Sciences (IES), "Active Learning at Scale: Transforming Teaching and Learning via Large-Scale Learning Science and Generative AI" Role: Co-PI (PI: Danielle McNamara). $3,750,000 total; local $210,000 2024-2027

National Science Foundation (NSF), "INTERACT: An Incubator to Enable Scalable Education Equity Research with Terracotta" Role: PI (joint with Mary Murphy; Co-PIs: Amanda Diekman, Emily Fyfe, Rob Goldstone, Dorainne Green). $499,809 2023-2025

XPRIZE, Digital Learning Challenge, Finalist ($83,333) + Runner-Up ($125,000) prizes. "Terracotta" Role: PI. 2022

Institute of Education Sciences (IES), "The Canvas+Terracotta LMS-Based Experimental Education Research Platform" Role: PI (Co-PI: Mark McDaniel). $2,000,000 2021-2027

Reboot Foundation, "Improving Critical Thinking with a Categorization Practice Intervention" Role: PI (joint with Emily Fyfe). $5,000 2020

Schmidt Futures, "Rapid Experimentation Educators Network" Role: PI (Co-PI: Ulrich Boser). $450,000 2020

Unizin, "Social Influence from Peer Assessment of College Coursework" Role: PI (Co-PIs: Matt Baldwin, Rob Goldstone). $20,000 2020

Schmidt Futures, "COVID-19 and Perceptions of Learning Technology" Role: PI. $45,000 2020

Google, Google Cloud COVID-19 Research Credits Program, "Behavioral Correlates of LMS Engagement" Role: PI. $1,410 2020

Indiana University, Office of Research Administration "Crisis Transition to Online Learning" Role: PI. $40,000 2020

Indiana University, Institute for Advanced Study, "Provocations: Conversations Towards a Bold University in the 21st Century" Role: PI. $4,950 2019

Indiana University, SoTL Grant, "Improving Teaching Through Enhanced Analysis of Teaching Evaluations" Role: Co-PI (PI: Richard Hullinger). $5,000 2017

Association for Psychological Science (APS), APS Fund for Teaching and Public Understanding of Psychological Science, "Improving Student Motivation and Success with Social Norm Messaging" Role: PI (Co-PIs: Ed Hirt, David Landy, and Rob Goldstone). $11,000 2016

Bay View Alliance, "Charting Educational Outcomes Following P101 with Linear Modeling: Enabling Data-Driven Improvement of Introductory Psychology" Role: PI (Co-PIs: David Landy and Tom Busey). $30,000 — 2014

Indiana University, CITL Summer Writing-Teaching Grant, Role: PI. $1,500 — 2012

Indiana University, SoTL Active Learning Grant, "Making Public Service Announcements as a Teaching Aid for Cognitive Psychology" Role: PI. $1,500 — 2009

Publications

Journal Articles

*ˆ indicates advisee co-author; * indicates co-first authors*

Motz, B.*, Hullinger, R.*, Celestin, B., Butz, H.ˆ, & Kruschke, J. (2026). Differences of opinions: How visualizations of uncertainty in ratings data affect choice. *Applied Cognitive Psychology, 40*, e70213. 10.1002/acp.70213

Motz, B., Magliano, J., McCarthy, K., & McNamara, D. (2026). What understanding is for. *Computational Brain and Behavior*. 10.1007/s42113-026-00300-z

Attari, S.*, **Motz, B.***, Mishra, A., Brautigam, G., Trapp, T., & Graham, J. (2026). Do facts matter? Consumer misperceptions about adopting electrification technologies. *Climatic Change, 179*, 35. 10.1007/s10584-026-04120-9

Rust, M., & **Motz, B.** (2026). Incorporating an LMS learning analytic into proactive advising: Validity and use in a randomized experiment. *Internet and Higher Education, 68*, 101057. 10.1016/j.iheduc.2025.101057

Narwal, A.ˆ, Fyfe, E., & **Motz, B.** (2025). Simulation and symbolic thinking in equations representing change. *PLOS One, 20*(12), e0339004. 10.1371/journal.pone.0339004

Motz, B., Diekman, A., Goldstone, R., Green, D. J., Fyfe, E., Bernardini, S., Emerson, K., Jankowski, H., Ansari, S., Arney, M., Barbieri, C., Baumgartner, E., Beam, M., Boucher, K., Callison, M., Canning, E., Chen, X.-Y., Chow, J., Clark, T., Gensic, J., Godwin, A., Gök, S., Gunderson, E., Kung, F., Laski, E., Mak, J., Master, A., Matthews, P., Merrick, M., Narwal, A.ˆ, Newkirk, B., Olszewski, B., Rice, A., Shelby, M., Sidney, P., Snapp-Childs, W., Stewart, M., Thompson, C., Tipton, E., Vuletich, H., White, A., Winegar, A., Yan, V., Zepeda, C., Zhang, T., & Murphy, M. C. (2025). A field-initiated vision of research infrastructure for STEM education. *International Journal of STEM Education Research*. 10.1186/s40594-025-00581-z

Motz, B., Chinni, A.ˆ, de Leeuw, J., Jankowski, H., Aggarwal, A., Amato, M., Berlin, K., Britten, K., Brown, A., Cerchiaro, M., Evans, N., Findley, A., Gorman, R., Gregg, K., Hansen, K., Hollender, H., Hullinger, R., Larkin, P., Lion, M., Long, R., Mannarino, A., Mocko, M., Moore, K.ˆ, Packowski, J., Palmer, C., Ritchie, K., Scott, J., Stanton, M., Talcott, L., Wagner, L., Waite, M., Yeager, R., and Fyfe, E. (2025). ManyClasses 2: The effects of prequestions on media interactions and learning, *Journal of Educational Psychology*. 10.1037/edu0000978

Gigerenzer, G., Allen, C., Gaillard, S., Goldstone, R., Haaf, J., Holmes, W., Kashima, Y., **Motz, B.**, Musslick, S., & Stefan, A. (2025). Alternative models of funding curiosity-driven research. *Proceedings of the National Academy of Sciences, 122*(5). 10.1073/pnas.2401237121

Motz, B. (2024). Concentration toward the mode: Estimating changes in the shape of a distribution of student data. *Journal of School Psychology, 107*, 101364. 10.1016/j.jsp.2024.101364

Motz, B., Üner, Ö, Jankowski, H., Christie, M., Burgas, K., del Blanco Orobitg, D., & McDaniel, M. (2024). Terracotta: A tool for conducting experimental research on student learning. *Behavior Research Methods, 56*, 2519-2536. 10.3758/s13428-023-02164-8

Lu, X., Wang, W., **Motz, B.**, Ye, W., & Heffernan, N. (2023). Immediate text-based feedback timing on foreign language online assignments: How immediate should immediate feedback be? *Computers and Education Open, 5*(15), 100148. 10.1016/j.caeo.2023.100148

Motz, B., Bergner, Y., Brooks, C., Gladden, A.ˆ, Gray, G., Lang, C., Li, W., Marmolejo-Ramos, F., & Quick, J. (2023). A LAK of direction: Misalignment between the goals of learning analytics and its research scholarship. *Journal of Learning Analytics, 10*(1), 1-13. 10.18608/jla.2023.7913

Motz, B., & Morrone, A. (2023). Wild brooms and learning analytics. *Journal of Computing in Higher Education*. 10.1007/s12528-023-09353-6

Motz, B., Fyfe, E., & Guba, T. (2022). Learning to call bullsh*t via induction: Categorization training improves critical thinking performance. *Journal of Applied Research in Memory and Cognition*, https://doi.org/10.1037/mac0000053

de Leeuw, J.*, **Motz, B.***, Fyfe, E., Carvalho, P., & Goldstone, R. (2022). Generalizability, transferability, and the practice-to-practice gap. Commentary in response to T. Yarkoni, The Generalizability Crisis. *Behavioral and Brain Sciences, 45*, e11. 10.1017/S0140525X21000406

Motz, B., Quick, J., & Morrone, A. (2022). When online courses became the student union: Utilization of learning technologies for peer interaction during COVID-19. *Technology, Mind, and Behavior, 3*(1). 10.1037/tmb0000061

Motz, B., Goldstone, R., Busey, T., & Prather, R. (2021). Visual search asymmetry due to the relative magnitude represented by number symbols. *Vision, 5*(3), 42. 10.3390/vision5030042

Motz, B., Canning, E., Green, D., Mallon, M., & Quick, J. (2021). The influence of automated praise on behavior and performance. *Technology, Mind, and Behavior, 2*(3). 10.1037/tmb0000042

Fyfe, E., de Leeuw, J. R., Carvalho, P. F., Goldstone, R., Sherman, J.ˆ, Admiraal, D., Alford, L. K., Bonner, A., Brassil, C. F., Brooks, C. A., Carbonetto, T., Chang, S. H., Cruz, L., Czymoniewicz-Klippel, M., Daniel, F., Driessen, M., Habashy, N., Hanson-Bradley, C. L., Hirt, E. R., Carbonell, V. H., Jackson, D. K., Jones, S., Keagy, J. L., Keith, B., Malmquist, S. J., McQuarrie, B., Metzger, K. J., Min, M. K., Patil, S., Patrick, R. S., Pelaprat, E., Petunich-Rutherford, M., Porter, M., Prescott, K., Reck, C., Renner, T., Robbins, E., Smith, A. R., Stuczynski, P., Thompson, J., Tsotakos, N., Turk, J. K., Unruh, K., Webb, J. D., Whitehead, S. N., Wisniewski, E. C., Zhang, K. A., & **Motz, B.** (2021). ManyClasses 1: Assessing the generalizable effect of immediate versus delayed feedback across many college classes. *Advances in Methods and Practices in Psychological Science, 4*(3), 1-24. 10.1177/25152459211027575

Motz, B., Mallon, M., & Quick, J. (2021). Automated educative nudges to reduce missed assignments in college. *IEEE Transactions on Learning Technologies, 14*(2), 189-200. 10.1109/TLT.2021.3064613

Motz, B., Quick, J., Wernert, J., & Miles, T. (2021). A pandemic of busywork: Increased online coursework following the transition to remote instruction is associated with reduced academic achievement. *Online Learning, 25*(1), 70-85. 10.24059/olj.v25i1.2475

Andrews, C.ˆ, **Motz, B.**, Israel, J., & Leary, H. (2020). Characteristics of students who opted in to use the Boost mobile app as an educational support service. *Journal of Teaching and Learning with Technology, 9*(1), 158-170. 10.14434/jotlt.v9i1.29376

Eyink, J.*ˆ, **Motz, B.***, Heltzel, G.ˆ, & Liddell, T. (2020). Self-regulated studying behavior, and the social norms that influence it. *Journal of Applied Social Psychology, 50*(1), 10-21. 10.1111/jasp.12637

Motz, B., Carvalho, P., de Leeuw, J., & Goldstone, R. (2018). Embedding experiments: Staking causal inference in authentic educational contexts. *Journal of Learning Analytics, 5*(2), 47-59. 10.18608/jla.2018.52.4

Motz, B., de Leeuw, J., Carvalho, P., Liang, K.ˆ, & Goldstone, R. (2017). A dissociation between engagement and learning: Enthusiastic instructions fail to reliably improve performance on a memory task. *PLoS ONE, 12*(7): e0181775. 10.1371/journal.pone.0181775
(Winner of the Center for Open Science Preregistration Challenge Award)

de Leeuw, J. & **Motz, B.** (2016). Psychophysics in a Web browser? Comparing response times collected with JavaScript and Psychophysics Toolbox in a visual search task. *Behavior Research Methods, 48*(1), 1-12. 10.3758/s13428-015-0567-2

Carvalho, P., Braithwaite, D., de Leeuw, J., **Motz, B.**, & Goldstone, R. (2016). An in-vivo study of self-regulated study sequencing in Introductory Psychology courses. *PLoS ONE 11*(3): e0152115. 10.1371/journal.pone.0152115

Day, S., **Motz, B.**, Goldstone, R. (2015). The cognitive costs of context: The effects of concreteness and immersiveness in instructional examples. *Frontiers in Psychology, 6*(1876). 10.3389/fpsyg.2015.01876

Motz, B., Teague, J., & Shepard, L. (2015). Know thy students: An institutional report that provides aggregate student data to instructors. *EDUCAUSE Review Online (Peer Reviewed Category)*. http://www.educause.edu/ero/article/know-thy-students-providing-aggregate-student-data-instructors

Motz, B. (2013). Cognitive science in popular film: The Cognitive Science Movie Index. *Trends in Cognitive Sciences, 17*(10), 483-485. 10.1016/j.tics.2013.08.002

Motz, B., Erickson, M. & Hetrick, W. (2013). To the beat of your own drum: Cortical regularization of non-integer ratio rhythms toward metrical patterns. *Brain and Cognition, 81*(3), 329-336. 10.1016/j.bandc.2013.01.005

Motz, B., James, K. & Busey, T. (2012). The Lateralizer: A tool for students to explore the divided brain. *Advances in Physiology Education, 36*(3), 220-225. 10.1152/advan.00060.2012

Motz, B., Goldstein, M. & Smith, L. (2012). Understanding behavior from the ground up: Constructing robots to reveal simple mechanisms underlying complex behavior. *Psychology Learning and Teaching, 11*(1), 77-86. 10.2304/plat.2012.11.1.77

Núñez, R., **Motz, B.**, & Teuscher, U. (2006). Time after time: The psychological reality of the Ego- and Time-Reference-Point distinction in metaphorical construals of time. *Metaphor and Symbol, 21*(3), 133-146. 10.1207/s15327868ms2103_1

Motz, B., & Alberts, J. (2005). The validity and utility of geotaxis in infant rats. *Neurotoxicology and Teratology, 27*, 529-533 and reply 543-544. 10.1016/j.ntt.2005.06.005

Alberts, J., **Motz, B.**, & Schank, J. (2004). Positive geotaxis in infant rats: A natural behavior and a historical correction. *Journal of Comparative Psychology, 118*(2), 123-132. 10.1037/0735-7036.118.2.123

Under Review

Motz, B., Chinni, A.^, Barriball, A.^, & McNamara, D. (under review). Some assembly required: Self-testing alone is not an effective way to build broader understanding. https://doi.org/10.31234/osf.io/73mbh_v2

Motz, B., Moore, K.^, Gregg, K., & Roozenbeek, J. (under review). Prebunking via induction: Categorization practice reduces misinformation susceptibility. https://doi.org/10.31234/osf.io/w2bmp_v1

Other

Jaggars, S. **Motz, B.**, Rivera, M., Heckler, A., Quick, J., Hance, E., Karwisch, C. (2021). Digital divides at the University: Lessons learned from the COVID-19 Emergency Transition. Aaron Horn, Ed., *Midwest Higher Education Compact (MHEC)*. Minneapolis, MN.

Motz, B. (August 23, 2019). Principles for the responsible design of automated student support. *EDUCAUSE Review, Transforming Higher Ed Blog*. https://er.educause.edu/blogs/2019/8/principles-for-the-responsible-design-of-automated-student-support

Kiang, M., & **Motz, B.** (2005). Letter from the editors: Intersection Point. *Cognitive Science Online, 3*(2), i-ii. paper_pdf

Refereed Conference Papers

Motz, B., Chinni, A., Barriball, A., & McNamara, D. (2025). Some assembly required: Learning facts in isolation limits inferences. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society, 47*. https://escholarship.org/uc/item/3m55z785

Narwal, A., Goldstone, R., Fyfe, E., & Motz, B. (2025). Second hand effects: Exploring spatial influences on temporal judgments in clocks. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society, 47*. https://escholarship.org/uc/item/9p61c193

Gatto, O., Yong, C., Sirajuddin, A., & Motz, B. (2025). When rules don't cut it: The relative frequency of inductive and deductive language during real-world surgical training. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society, 47*. https://escholarship.org/uc/item/3gc1f7fm

Fan, J., Zheng, K., Motz, B., Doroudi, S., Son, J., & Thille, C. (2025). Minds at school: Advancing cognitive science by measuring and modeling human learning in situ. In *Proceedings of the 47th Annual Conference of the Cognitive Science Society*.

Motz, B., Jankowski, H., Lopatin, J., Tseng, W., & Tate, T. (2024). Using a platform to run an experiment outside the platform. In *Fifth Annual Workshop on A/B Testing and Platform-Enabled Learning Research*, companion proceedings of the 11th ACM Conference on Learning @ Scale.

Ritter, S., Fancsali, S., & Murphy, A., Heffernan, N., Motz, B., Mallick, D.B., Roschelle, J., McNamara, D., & Williams, J.J. (2024). Fifth Annual Workshop on A/B Testing and Platform-Enabled Learning Research. In *Proceedings of the 11th ACM Conference on Learning @ Scale*. 10.1145/3657604.3664642

Motz, B. & Jankowski, H. (2023). Embedding experimental research in the learning management system. Division D – Measurement and Research Methodology. *2023 Conference of the American Educational Research Association (AERA)*.

Ritter, S., Heffernan, N., Williams, J.J., Lomas, D., Bicknell, K., Roschelle, J., Motz, B., McNamara, D., Baraniuk, R., Basu Mallick, D. and Kizilcec, R. (2023). Fourth Annual Workshop on A/B Testing and Platform-Enabled Learning Research. In *Proceedings of the 10th ACM Conference on Learning@Scale*. 10.1145/3573051.3593397

Quick, J., Motz, B., & Morrone, A. (2023). Lost in translation: Determining the generalizability of temporal models across course contexts. In *Proceedings of the 13th International Conference on Learning Analytics & Knowledge (LAK23)*. 10.1145/3576050.3576092

Ritter, S., Heffernan, N., Williams, J. J., Bicknell, K., Lomas, D., Roschelle, J., Motz, B., McNamara, D., Baraniuk, R., Mallick, D., Kizilcec, R., & Baker, R. (2022). Third Annual Workshop on A/B Testing and Platform-Enabled Learning Research. In *Proceedings of the 8th ACM Conference on Learning @ Scale.* 10.1145/3491140.3528288

Motz, B., Brooks, C., Quick, J., Bergner, Y., Gray, G., Lang, C., Li, W., Marmolejo-Ramos, F. (2022). A baseline measure of open research practices in learning analytics. In *Building Open Science and Scholarship in SoLAR*, a workshop associated with the *12th Learning Analytics and Knowledge Conference*. 10.35542/osf.io/325d7

Motz, B., Carvalho, P., & Fyfe, E. (2020). A preliminary taxonomy of A/B: Education experiments with different inferences and scopes. In *Educational A/B Testing at Scale*, a workshop associated with the *7th ACM Conference on Learning @ Scale*.

Quick, J., Motz, B., Israel, J., & Kaetzel, J. (2020). What college students say, and what they do: Aligning self-regulated learning theory with behavioral logs. In *Proceedings of the 10th International Conference on Learning Analytics & Knowledge (LAK20)*. 10.1145/3375462.3375516

Fyfe, E. R., de Leeuw, J. R., Carvalho, P. F., Goldstone, R. L., Sherman, J., & Motz, B. A. (2020, April). Large-scale collaborative science: The ManyClasses approach to experimental educational research. In S. J. Peters (chair), Actions for increasing the credibility of educational research. Symposium accepted at the *American Education Research Association (AERA)*, San Francisco, CA. (Conference canceled). 10.35542/osf.io/tv7rn

Motz, B., Quick, J., Schroeder, N., Zook, J., & Gunkel, M. (2019). The validity and utility of activity logs as a measure of student engagement. In *Proceedings of the 9th International Conference on Learning Analytics & Knowledge (LAK19)*. 10.1145/3303772.3303789

Motz, B. & Carvalho, P. (2019). Not whether, but where: Scaling-up how we think about effects and relationships in natural educational contexts. In *Companion Proceedings of the 9th International Conference on Learning Analytics & Knowledge (LAK19)*. 10.13140/RG.2.2.30825.34407

Motz, B., Busey, T., Rickert, M., Landy, D. (2018). Finding topics in enrollment data. In *Proceedings of the 11th International Conference on Educational Data Mining*. Buffalo, New York. https://eric.ed.gov/?id=ED593218

Carvalho, P., Gao, M., Motz, B., & Koedinger, K. (2018). Analyzing the relative learning benefits of completing required activities and optional readings in online courses. In *Proceedings of the 11th International Conference on Educational Data Mining*. Buffalo, New York. https://eric.ed.gov/?id=ED593230

de Leeuw, J., Motz, B., Eastwood, J., Maltese, A., Goldstone, R., & Danish, J. (2015). Needle in the neural haystack: EEG signatures of concept learning while viewing naturalistic educational materials. Paper presented at the 2015 annual meeting of the *American Educational Research Association*, Chicago, Illinois.

Carvalho, P.F., Braithwaite, D. W., de Leeuw, J. R., Motz, B. A., & Goldstone, R.L. (2014). Effectiveness of Learner-Regulated Study Sequence: An in-vivo study in Introductory Psychology courses. In *Proceedings of the 37th Annual Conference of the Cognitive Science Society*.

Motz, B. (2013). Fantasy football: A touchdown for undergraduate statistics education. Paper and presentation at the annual meeting of *Games+Learning+Society*, Madison, Wisconsin.

Motz, B., & Núñez, R. (2004). Implications of new baseline data in the spatial construal of time. In *Proceedings of the 26th Annual Conference of the Cognitive Science Society*.

Invited Talks, Seminar Presentations, Posters, and Guest Lectures

20 Jul 2026 InstructureCon Partner Summit 2026. Louisville, KY. Plenary panel: *Evidence, Outcomes, and Trust*.

22 Apr 2026 Department of Learning Sciences. Georgia State University, Atlanta, GA. Invited Presentation.

5 Mar 2026 University of Texas Arlington. Arlington, TX. Guest Online Lecture to EDUC 5360 Introduction to Mind, Brain, and Education. Organizer: Dan Robinson

4 Feb 2026 LERN 2026, Arizona State University. Tempe, AZ. Invited plenary panelist: *Taking Human-Centered Learning Innovation to Scale*. (McNamara, D., Ritter, S., Wise, A., Heffernan, C., Motz, B.).

17 Dec 2025 SREE Methods Webinar: *From Clicks to Causal Claims: Methodological Innovations in Digital Learning Platform Research*. Invited Presentation. (Roschelle, J., Botelho, A., Motz, B.)

23 Nov 2025 Psychonomics 2025. Denver, CO. Presentation: *Teaching Students to Spot Misinformation: Does Learning-by-Doing Beat Learning-by-Knowing?* (Motz, B. & Gregg, K.).

21 Nov 2025 Psychonomics 2025. Denver, CO. Poster: *Examining Self-regulated Learning of Natural Science Categories through Adaptive Learning Systems* (Hu, M., Nosofsky, R., Motz, B.).

21 Nov 2025 Psychonomics 2025. Denver, CO. Symposium presentation: *Boundaries, Generalizability, and Translating Science to Educational Practice*.

11 Oct 2025 SREE 2025. Chicago, IL. Moderated discussion: *Classroom-Embedded Experiments at Scale: New Methods and Ethical Considerations for Digital Learning Platforms* (Roschelle, J., Wei, X., Closser, A., Basu Mallick, D., Motz, B., & Ritter, S.).

28 Jul 2025 Nightingale Labs 2025 Workshop, Stanford University. Palo Alto, CA. Invited Presentation.

17 Jun 2025 Program for Open Scholarship and Education (POSE). University of British Columbia. Invited Online Presentation.

03 Jun 2025 1EdTech Learning Impact Conference 2025. Indianapolis, IN. Panel: *Bringing Learner Data into the Research Lab*. (Motz, B., Basu Mallick, D., Cleary, A., & Hecht, C.)

14 May 2025 IU Learning Analytics Faculty Fellows (LAFF) Program. Indianapolis, IN. Keynote for Inaugural Kick-Off Meeting.

22 Jan 2025 Cognitive and MCP Colloquium, Department of Psychological Sciences, Purdue University, West Lafayette, IN. Invited Presentation.

2 Dec 2024 Innovation Insights Speaker Series, Center for Academic Innovation, University of Michigan. Invited Online Presentation.

15 Oct 2024 XPRIZE Alumni Trailblazers webinar series, XPRIZE.org. Invited Online Presentation.

21 Feb 2024 LERN Speaker Series, Learning Engineering Institute, Arizona State University. Tempe, AZ. Invited Presentation.

14 Feb 2024 University of Southern Indiana's 8th annual Celebration of Teaching and Learning Symposium. Evansville, IN. Keynote Presentation.

13 Feb 2024 SoTL Workshop, Indiana University Center for Innovative Teaching and Learning. Bloomington, IN.

18 Nov 2023 Psychonomics 2023. San Francisco, CA. Presentation: *Differences of Opinions: How Visualizations of Uncertainty in Ratings Data Affect Choice*. (Motz, B., Hullinger, R., Celestin, B., Butz, H., & Kruschke, J.).

17 Nov 2023 Psychonomics 2023. San Francisco, CA. Poster: *Simulation and Symbolic Thinking in Equations Representing Change* (Narwal, A., Motz, B., & Fyfe, E.).

16 Nov 2023 International Association of Metacognition. San Francisco, CA. Presentation: *Preview of ManyClasses 2: Assessing the Effect of Pretesting on Student Learning and Behavior Across Diverse Settings and Materials.*

25 Oct 2023 *INTERACT Incubator*. Organizer of 3-day convening of the INTERACT Incubator. Bloomington, IN.

21 Sep 2023 IES Innovation Day. Washington, DC. Invited Panel (Stephenson, S. P., Heffernan, N., Motz, B., Ritter, S., Mallick, D., & Davenport, J.).

26 Aug 2023 Visual Science of Art Conference (VSAC). Nicosia, Cyprus. Poster: *Using Instruction to Alter Fixation Patterns in Abstract Art.* (Chinni, A. & Motz, B.).

21 Jul 2023 Program for Open Scholarship and Education (POSE). University of British Columbia. Invited Online Presentation.

7 Jul 2023 Learning Impact Conference 2023, Anaheim, CA. Invited Presentation.

13 Apr 2023 AERA 2023, Chicago, Illinois. Poster: *Embedding Experimental Research in the Learning Management System.* (Motz, B., Jankowski, H.).

20 Nov 2022 Psychonomics 2022, Boston, MA. Presentation: *Terracotta: A Tool for Conducting Experimental Research in Classrooms on Student Learning.* (Motz, B., Jankowski, H., & McDaniel, M.).

19 Nov 2022 Psychonomics 2022. Boston, MA. Poster: *An Inductive Approach to Improving Critical Thinking.* (Guba, T., Fyfe, E., & Motz, B.).

4 Nov 2022 AAC&U Transforming STEM Higher Education Conference. Arlington, Virginia. Presentation: *Experimental Research on STEM Education with Terracotta.*

28 Apr 2022 Unizin Summit. Online Keynote Presentation.

2 April 2022 Learning & the Brain, *The Science of Knowledge,* New York, NY. Invited speaker.

12 Oct 2021 EDUCAUSE Industry and Campus Solutions Webinar.

29 Sep 2021 PEERS Data Hub Webinar, co-sponsored by AERA and ICPSR. Invited Online Presentation.

22 Sep 2021 OLC Accelerate 2021 Research Summit. Invited Online Presentation.

16 Jun 2021 CASE 2021. Online. Invited plenary co-discussant with Kumar Garg.

21 Apr 2021 2021 Unizin Summit. Online. Presentation: *ManyClasses 1: An Unprecedented Learning Experiment – Findings, Lessons Learned, and Looking Ahead*.

29 Mar 2021 Methodology SIG of the Society for Learning Analytics. Co-organizer of online discussion panel, *Should your (quantitative) methods change if you care about social justice?*

9 Feb 2021 AAAI Conference on Artificial Intelligence. Online. Invited panelist at workshop, *Imagining Post-COVID Education with AI*.

26 Feb 2021 Florida Distance Learning Research Consortium's monthly webinar series. Invited Online Presentation.

3 Dec 2020 International Society for Technology in Education (ISTE Live). Featured Online Presentation.

18 Nov 2020 Indiana University Institute for Advanced Study Bicentennial Symposium Series. Organizer: *eLearning Research Symposium*.

18 Sep 2020 Worcester Polytechnic Institute. Worcester, MA. Guest Online Lecture to CS525/565 Topics in Computer Science and SEME565 User Modeling. Seminar organizer: Neil Heffernan.

May 2020 Annual Meeting of the Association for Psychological Science, Chicago, IL. Symposium (Conference canceled).

20 Feb 2020 Unizin Hackathon, Research Track. Austin, Texas. Co-organizer.

15 Oct 2019 EDUCAUSE *Under the Ed Tech Radar Pitch Competition*. Chicago, Illinois. Invited Presentation.

25 Sep 2019 University of Minnesota, Learning Analytics Meet Up Series. Online Presentation at Inaugural Symposium.

19 Apr 2019 Unizin Summit. Denver, CO. Presentation: *Nudges as behavioral guide rails: An intervention system to proactively prevent students from missing assignments.*

17 Apr 2019 Unizin Summit. Denver, CO. Presentation: *ManyClasses v1: A call for participation in a game-changing experiment on the psychology of learning.* (Motz, B. & Fyfe, E.)

15 Jul 2018 replicate.education: A Workshop on Large Scale Education Replication. Buffalo, NY. Presentation: *ManyClasses: A model for abstracting generalizable research principles from different learning contexts.*

1 Mar 2019 School for Public and Environmental Affairs (SPEA) Connect 2019 Teaching Workshop. Bloomington, Indiana. Invited Keynote Presentation.

15 Nov 2018 International Association of Metacognition. New Orleans, LA. Presentation: *Rendering retrieval practice as precious or painful.*

10 Nov 2017 IU Online Conference. Indianapolis, IN. Presentation: *Online self-directed learning activities, and the norms that influence them.*

29 Sep 2017 Indiana Students and Teachers of Psychological Science (ISTOPS) 2017 Conference. Marian University, Indiana. Invited Keynote Presentation.

8 Aug 2017 Primary Source Immersion Program, IU Libraries. Bloomington, Indiana. Invited Keynote Presentation.

26 Jul 2017 Indiana University's Center for Innovative Teaching and Learning. Bloomington, Indiana. Invited workshop presentation and accompanying webinar.

9 Jul 2017 Beyond the Lab: Using Big Data to Discover Principles of Cognition, Leading Edge Workshop of the Psychonomic Society, Madison, WI. Poster: *Big data: An exploratory study of in-vivo online courses.* (Yu, J., Motz, B., & Landy, D.).

25 Feb 2017 Annual Meeting of the Council of Graduate Departments of Psychology (COGDOP). Greenville, SC. Invited Presentation.

25 May 2016 Society for the Teaching of Psychology Teaching Institute at the 23rd Annual Conference of the Association for Psychological Science, Chicago, IL. Poster: *Educational outcomes following enrollment in Introductory Psychology*. (Motz, B., & Rickert, M.).

1 Apr 2016 Indiana University, Center for Innovative Teaching and Learning Celebration of Teaching 2016. Bloomington, IN. Master of ceremonies and co-organizer for inaugural teaching hackathon.

12 Feb 2016 Preparing Future Faculty Conference. Bloomington, Indiana. Invited Panelist.

4 Feb 2016 Hoosier Association for Science Teachers, Inc. (HASTI). Indianapolis, IN. Presentation: *Positioning Psychology Among Indiana's High School Sciences.*

21 Oct 2015 Indiana University's Statewide IT 2015 Conference. Bloomington, IN. Presentation: *Ubiquitous lightweight assessment in an online course*.

27 Feb 2015 Indiana University's Health Professions and Prelaw Center (HPPLC). Bloomington, IN. Seminar presentation.

8 Oct 2014 Indiana University's Statewide IT 2014 Conference. Bloomington, IN. Presentation: *Know thy students: An institutional report on aggregate student data*.

6 Feb 2014 Hoosier Association for Science Teachers, Inc. (HASTI). Indianapolis, IN. Presentation: *The Psychological Science: Mind, Brain, & Behavior.*

11 Oct 2013 125th anniversary celebration of the Department of Psychological and Brain Sciences. Bloomington, IN. Invited moderator for plenary discussion panel: *Psychology in the Real World*.

8 Feb 2013 University of Louisville Delphi Celebration of Teaching & Learning. Louisville, Kentucky. Invited Plenary Presentation.

5 Apr 2011 Indiana University Center for Innovative Teaching and Learning, Spotlight on Innovation, Bloomington, IN. Poster: *Student-generated public service announcements: A case study of the need for careful assessment of learning outcomes.*

21 Oct 2010 Society for Psychophysiological Research, Portland, OR. Poster: *Rhythm of perception and attention: An OEP study of temporal attention*. (Erickson, M., Motz, B., & Hetrick, W.).

15 Jun 2005 Society for Music Perception and Cognition, San Diego, CA. Presentation: *Effects of tempo on rhythmic expectations*. (Motz, B., Urbach, T., & Kutas, M.).

2001 International Society for Developmental Psychobiology, San Diego, CA. Poster: *Positive evidence of geotaxis in 10-day-old rats*. (Motz, B., & Alberts, J.)

Appointments

Associate Director – *2022 - current*
Director – *2019 - 2022*
eLearning Research and Practice Lab, Pervasive Technology Institute

Faculty Fellow for Academic Analytics, eDS, University Information Technology Services – *2018 - 2020*

Director of Undergraduate Instruction and Online Development – *2015-2018*
Director of Pedagogy – *2012-2014*
IU Department of Psychological and Brain Sciences

Service

Guest-Editor, Contemporary Educational Psychology – *2026*
Emerging Shared Research Infrastructure in Educational Psychology

Selected Participant, Rural Indiana Faculty Tour, IU Center for Rural Engagement – *2026*

Editorial Board Member, Journal of Applied Research in Memory and Cognition – *2026-current*

Elected Member, Space Committee, IU Department of Psychological and Brain Sciences – *2023-current*

Spokesperson, Cognitive Area, IU Department of Psychological and Brain Sciences – *2023-2026*
Appointee, Graduate Program Committee, Graduate Admissions Committee

Organizing Committee, Methodology in Learning Analytics SIG, SoLAR – *2020-2024*

Lead Partner, Charting the Future, Indiana University, University Academic Affairs – *2021-2023*

Invited Member, IUB2030 Strategic Planning Working Group, Indiana University Bloomington – *2022*

Invited Member, Research Committee, Unizin Consortium – *2019-2021*

Academic Editor, PLoS ONE – *2018-2022*

Undergraduate Curriculum Committee, IU Department of Psychological and Brain Sciences – *2009-2021*
Chair, Ad Hoc Assessment Committee – *2013-2014*

Curator, Designer, and Developer, Cognitive Science Movie Index – *2010-2018*

Invited Member, Educational Policies Committee, IU Bloomington Faculty Council – *2014-2016*

Active Learning Online Faculty Learning Community, IU CITL – *2015-2016*

Invited Member, Indiana University Faculty Leadership Institute, IU FACET – *2014-2016*

Elected Member, Student Academic Appointee Board of Review, IU Bloomington Faculty Council – *2014-5*

Consultant and On-Camera Interviewee, NFL Network – *2011-2014*

Organizer, Pedagogy Seminar, IU Department of Psychological and Brain Sciences – *2011-2014*

Preparing Future Professors Faculty Learning Community, IU CITL – *2010-2014*

Invited Member, Undergraduate Strategic Planning Committee, Indiana University Bloomington – *2013*

Branding and Website Committee, IU Department of Psychological and Brain Sciences – *2009-2013*
Principal designer and developer of departmental website (2011 redesign)

Co-Editor – UCSD Cognitive Science Online, *2004-2005*

Advisory Boards: IES R305N240063 (PI: Akira Miyake), NSF-DUE 2417328 (PI: Jennifer Knight), NSF-DRL 1713567 (PI: Judith Fan), NSF-DRL 2243530 (PI: David Yeager)

Reviewer for: *Advances in Methods and Practices in Psychological Science; Behavioral Sciences; Biological Psychology; British Journal of Educational Psychology; British Journal of Educational Technology; Cognition and Instruction; Educational Psychology Review; Human Brain Mapping; IEEE Transactions on Neural Systems and Engineering; Journal of Applied Research on Memory and Cognition; Journal of Educational Psychology; Journal of Quantitative Analysis in Sports; Journal of Teaching and Learning; Learning and Instruction; NeuroImage; Oxford University Press; Perspectives on Psychological Science; PLoS ONE*; *Psychonomic Bulletin & Review; Technology, Mind, and Behavior*

Courses Taught

[Hybrid] From Inquiry to Insight: Measurement and Analysis Skills in Psychological and Brain Sciences (IU) – *2026-current*; Elementary data analysis and interpretation

Cognitive Psychology (IU) – *2009-currrent*
Upper-level survey of mental functions, theories, and experiments in cognitive psychology

Laboratory in Human Learning and Cognition (IU) – *2023-current*
Senior capstone course exploring experimental research on human learning and memory

Human Memory (IU) – *2022-current*
Upper-level overview of cognitive theories of human memory and associated processes

Introductory Data Analysis for Psychological and Brain Sciences (IU) – *2020-current*
Elementary data analysis methods including manipulating, summarizing, and visualizing data

[Online] Introductory Psychology 1 (IU) – *2015-2020*
Research methods, neuroscience, perception, learning, and cognitive psychology

Introduction to Psychological and Brain Sciences (IU) – *2009-2015*
Brain anatomy, perception, development, cognitive psychology, and social psychology

Prediction, Probability, and Pigskin (IU) – *2012-2014*
Freshman-level introduction to analytical techniques thru the lens of fantasy football

Methods in Experimental Psychology (IU) – *2009-2014*
Experimental methods, research design, and scientific writing

Teaching of Psychology (IU) – *2012-2014*
Graduate survey of pedagogical practices, emphasis on training lab instructors

General Psychology Honors (IU) – *2008-2012*
Survey of introductory psychology material, including weekly advanced labs and demonstrations

Neuroanatomy and Physiology (UCSD, with Jaime Pineda) – *2005*
Brain anatomy, cortical function, and neural information processing

Learning and Memory (UCSD, with Seana Coulson) – *2005*
Experimental foundations of psychological theories of learning and memory

Introduction to Cognitive Science (UCSD, with Jochen Triesch) – *2004*
Freshman course to provide interdisciplinary overview of Cognitive Sciences

Java Programming for Cognitive Modeling (UCSD, with John Batali) – *2004*
Introduction to Java programming and its applications in the modeling of cognitive processes

Design and Analysis of Experiments (UCSD, with John Batali) – *2004*
Mathematical introduction to probability, test design, and inferential statistics

Mathematical Foundations of Cognitive Science & Informatics (IU, with Ruth Eberle) – *2001-2002*
Introduction to mathematical and logical tools for building models in the information sciences

Society Memberships

Fellow, Psychonomic Society – *2018-current*
Member, Society for Applied Research in Memory and Cognition (SARMAC) – *2022-current*
Member, Cognitive Science Society – *2024-current*
Member, Association for Psychological Science (APS) – *2024-current*

Honors and Awards

Finalist & Runner Up – XPRIZE Digital Learning Challenge, 2023

Trustee Teaching Award – Indiana University, 2012, 2014, 2016, 2023

Tools Competition Mid-Range Prize – Futures Forum on Learning, 2021

Platinum Award (highest award), IMS Global Learning Impact Awards – IMS Global, 2019

President's Award for Excellence in Teaching and Learning Technology – Indiana University, 2015

Faculty Appreciation Award – IU Women's Basketball Program, 2009, 2014

Outstanding Teaching Award, Cognitive Science – Indiana University, 2013

Student Choice Award Nomination – Indiana University, 2009, 2010, 2011, 2012

Rising Star Award (annual office award) – Rapp Collins Los Angeles, 2008

Superior Teaching Award, Cognitive Science – UCSD, 2004, 2005, 2006

Outstanding Contribution Award, Cognitive Science – Indiana University, 2002

NIH Travel Award – National Institute of Health, 2001

Senior Leaders Award – Indiana University, 2001